

The six alkaline earth metals sit to the right of the alkali metals.

Alkaline Earth Metals

A slightly less reactive bunch than their group one neighbors, the group two elements are called “alkaline” because they form alkaline solutions when reacting with water. Soft and shiny when pure, they are called “earth metals” because many of them were discovered in their raw form in Earth’s crust. First purified in the 19th century, all alkaline earth metals are solid at room temperature. Each element in this group burns with a distinctive color.

Beryllium

4 This element is named after the mineral beryl.
Be Beryllium is released from the burning of fossil fuels and tobacco. This makes it a health hazard because it is chemically similar to magnesium and replaces it in the body, causing damage to cells. However, it has many applications when alloyed with other elements, such as in making missiles and satellites.



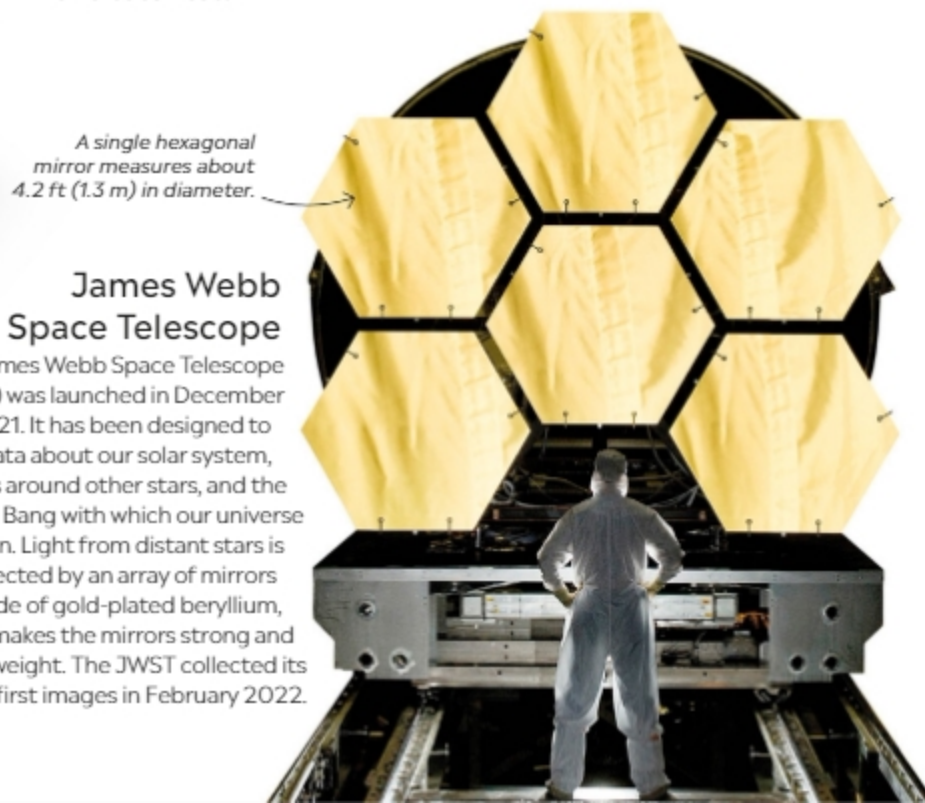
Aquamarine raw crystal

Aquamarine

One chief source of beryllium is beryl, a naturally occurring mineral made up of beryllium, aluminum, silicon, and oxygen. Beryl can take on many different colors, depending on the impurities present. Aquamarine is a type of blue beryl gemstone: the presence of iron makes it blue.

James Webb
Space Telescope

NASA's James Webb Space Telescope (JWST) was launched in December 2021. It has been designed to collect data about our solar system, planets around other stars, and the Big Bang with which our universe began. Light from distant stars is collected by an array of mirrors made of gold-plated beryllium, which makes the mirrors strong and lightweight. The JWST collected its first images in February 2022.



A single hexagonal mirror measures about 4.2 ft (1.3 m) in diameter.